Economics of Wildlife Governance: A Case Study of Alligators in The United States of America

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May 14th, 2025

Abstract

This paper explores the decline and conservation of the American Alligator (*Alligator Mississippiensis)* from the early 19th century to the present. It examines population trends across decades and analyzes how legal and illegal human practices affect the species. The study also investigates the economy surrounding alligators and how governance through state and federal laws influenced conservation efforts. The research uses secondary sources to draw on harvest and population data, including limited records of illegal poaching in states like Florida, which hint at an underground economy. Key findings highlight the major causes behind the species' decline, the profitability of the alligator trade across industries- fashion, military, infrastructure, and food- and the legislative responses that helped restore population levels. I conclude that the open-access model, which allowed unrestricted hunting without property rights or regulation, led to the overharvest and collapse of alligator populations. This collapse eventually prompted the development of governance systems that aligned economic incentives with conservation, enabling the sustainable recovery of the species.

Introduction

The American alligator (*Alligator mississippiensis*) is a large, semi-aquatic reptile native to the southeastern United States. It's normally found in warm freshwater environments like swamps, marshes, rivers, and lakes. Louisiana has the largest wild alligator population, followed by Florida, Georgia, Texas, South Carolina, Alabama, Mississippi, North Carolina, Arkansas, and Oklahoma in descending order of their alligator population. Alligators can grow between 6 and 14 feet long and are recognized by their dark coloration, armored bodies, and long snouts. Alligators have economic value due to the high demand for their leather and meat. Alligator leather is durable and uniquely textured, making it a popular material for luxury goods such as belts, shoes, wallets, and handbags. Additionally, alligator meat is considered a delicacy in parts of the Southern U.S. This demand created strong economic incentives for harvesting the species, especially in states with large populations.

During the early to mid-20th century, the commercial demand for alligator products led to overharvesting. Without regulation, alligator populations dropped drastically. This was a classic example of the "tragedy of the commons" since alligator hunting was unregulated in the country. In 1967, the alligator was officially listed as an endangered species under the Endangered Species Act due to the sharp decline in its numbers.

Government intervention played a key role in the survival of alligators. Strict regulations were introduced to ban unregulated hunting, and conservation programs focused on rebuilding populations. At the same time, states allowed controlled harvesting and established alligator farming industries, which created legal, sustainable ways to meet economic demand. These policies realigned incentives by making conservation economically beneficial and discouraging illegal poaching.

The American alligator's recovery demonstrates the effectiveness of government regulation and sustainable management in addressing overexploitation. Once endangered due to unregulated hunting driven by economic demand, the species has rebounded through conservation efforts and controlled harvesting programs. Today, alligator populations are stable, and their regulated use provides ongoing economic benefits while preserving ecological stability.

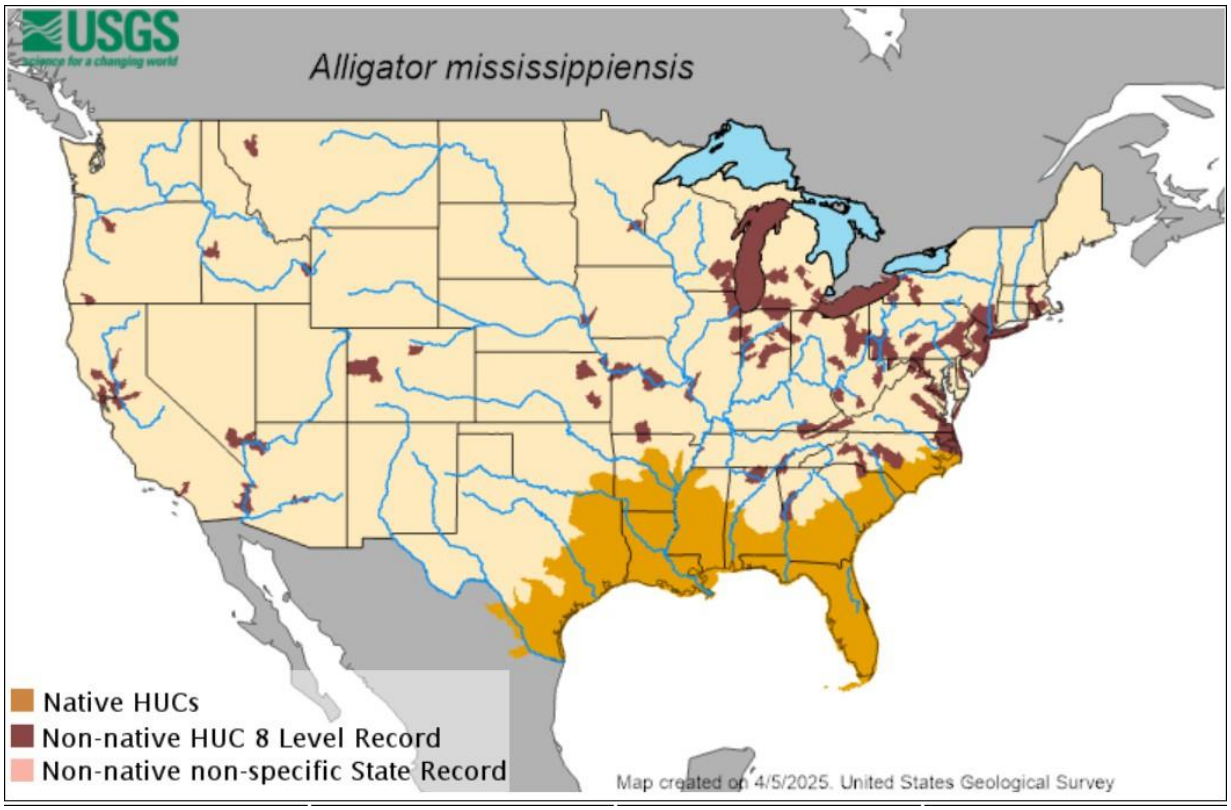


Figure 1: U.S. Geological Survey. 2025. Fact Sheet: Cyprinus carpio. Nonindigenous Aquatic Species Database

The figure above shows a map of the indigenous patterns of alligator habitation in the US. We can see that historically, alligators stuck to the southeastern coast of the US, shown in deep yellow on the map. The maroon depicts non-native regions of alligator sightings in the US, which tend to be near rivers.

The table below shows the timeline of the alligator population from the 1800s to the present.

Year	Population Estimate	Details
Early 1800s	3 million	The estimated population across the U.S. during the early 19th century.
Early 1900s	Thousands	Drastic decline due to unregulated hunting and habitat loss.
1967	Estimated to be < 100,000	Listed as an endangered species due to near-extinction.
1974	734,384 in Louisiana alone	Population estimate in Louisiana as recovery begins.
1987	2 million+	Removed from the endangered species list after significant recovery.
2000s	3 million+	U.S. population estimate after recovery, with substantial numbers in Florida.
Present	3–4 million+	The U.S. alligator population is currently estimated to exceed 3 million individuals.

Summary of Key Periods in American Alligator and Economic Significance

Before 1900, Native American tribes like the Seminole and Creek managed alligator use through subsistence hunting and communal land systems. Hunting was guided by cultural norms and remained sustainable due to limited market integration. The resource was not exploited commercially, and ecological balance was maintained through informal governance based on traditional practices and shared land use. These systems, though lacking formal enforcement, effectively prevented overharvest and supported continued use without causing population decline.

From 1900 to the 1940s, commercial hunting intensified under open-access conditions. Without regulation or property rights, rising hide prices led to overharvest, especially in Florida and Louisiana. The peak came in 1929, when Florida exported around 190,000 hides. Short-term profits were high, but long-term sustainability collapsed. Populations fell sharply due to unrestricted access, consistent with rent dissipation and biological decline predicted by resource economics. Governance was absent, and incentives favored extraction over conservation.

Between the 1940s and 1960s, alligator populations continued to decline, but most states took little regulatory action. The Lacey Act and Endangered Species Act were introduced federally, but enforcement was weak. The species lost economic value, and the absence of incentives discouraged conservation. Commercial activity decreased as populations dropped, but without clear rules or ownership, habitat degradation and illegal hunting persisted. This period was marked by regulatory inaction and continued to decline in both population and market presence.

In the 1970s and 1980s, state and federal governments introduced hunting bans and protection programs. Alligator populations began to recover as Louisiana and Florida monitored stocks and launched pilot ranching models. This command-and-control approach

restored ecological stability and reduced poaching. Although commercial use was still limited, regulation laid the foundation for future economic involvement. Conservation shifted from passive protection to active management, with enforcement and biological monitoring driving the recovery.

During the 1980s and 1990s, states moved toward market-based governance. Tag systems regulated harvests, and legal farming expanded. Louisiana's ranching model collected wild eggs, reared juveniles, and released a portion back into the wild. This allowed legal profit from alligator use while supporting population growth. Conservation outcomes improved as landowners, hunters, and farmers gained financial incentives. Economic value was reconnected to ecological sustainability, turning alligator use into a regulated, profitable activity.

Since the 2000s, alligator governance has combined regulated harvests, farming, and federal trade oversight. Programs like CITES tagging ensure legal trade and traceability. States issue permits based on population surveys, and landowners receive quotas tied to habitat quality. The industry generates tens of millions annually, supporting rural jobs and wetland conservation. Market mechanisms and regulatory tools work together to maintain sustainability while preserving access to global luxury markets for alligator products.

Native American Interaction

The economic relationship between Native American tribes, particularly the Seminole and Miccosukee, and alligators in the southeastern United States is deeply embedded in traditional lifeways, cultural symbolism, and later adaptations to colonial and modern pressures. Alligators were hunted for meat to provide sustenance, and their hide and bones for practical and ceremonial use. The meat was a high-protein food source, particularly valuable in the swampy ecosystems of Florida and the Everglades, while the tough, water-resistant hides were crafted into armor, drums, shields, and clothing (Hudson 1976; Weisman 1999). These materials also became commodities in trade networks, especially after European contact. Native communities exchanged alligator hides for tools, fabrics, and weapons through colonial outposts along the Gulf and Atlantic coasts (Goss 2010). In the early 20th century, the Seminole and Miccosukee tribes made alligator hunting into a public performance, alligator wrestling, which became a major tourist attraction. This was not merely entertainment; it was a form of cultural preservation and economic self-sufficiency at a time when Native livelihoods were increasingly threatened by external pressures (National Park Service 1995; The New Yorker 2023).

In contemporary times, tribal members continue to engage in regulated alligator hunting and the legal sale of alligator products such as leather goods, meat, and tourism-based activities. These practices are typically overseen in collaboration with state wildlife agencies and often incorporate conservation goals, reflecting both economic pragmatism and environmental ethics (U.S. Fish and Wildlife Service n.d.).Modern tribal enterprises, including alligator farms and wildlife parks, provide employment and educational platforms while maintaining sustainable species use.

Traditional landownership and governance are the foundation of the respective tribes' interaction with alligators. The Seminole and Miccosukee communities practiced communal land use, favoring collective ownership over private property (Goss 1995).Hunting and fishing territories, including those used for alligator harvesting, were typically demarcated using natural landmarks such as trees, rivers, or distinct geographic features. These boundaries were well understood within and among tribal groups, minimizing conflict and encouraging sustainable practices (Hudson 1976). Even when pressured by U.S. federal policies that imposed Western property concepts, many Native communities retained customary use patterns, asserting sovereignty and access to ancestral hunting grounds. This legacy continues to inform legal and political debates over land rights, especially in ecologically sensitive regions like the Everglades, where environmental management intersects with tribal sovereignty (Weisman 1999; Smithsonian Institution n.d.).

Displacement and Loss of Control

During the 1800s, most Choctaw and Creek communities were forcibly removed to Indian Territory (now Oklahoma), effectively cutting them off from alligator habitats. While some tribal members remained in the Southeast, they lost access to swamps and wetlands that became private property or state land. This shift dismantled Indigenous systems of resource governance and reduced their ability to sustainably manage alligator harvests.
The Seminole, by contrast, managed to remain in Florida, though they were pushed deep into the Everglades. They continued to hunt alligators both for subsistence and, increasingly, for trade.

By the late 1800s, Seminole hunters were participating in the hide trade, selling gator skins to merchants in Miami. This marked a significant shift from cultural harvesting to market-driven exploitation, influenced by poverty and pressure to adapt.

Transition to Open-Access and Overharvest

Having no clear legal control over the wetlands and no recognition of tribal governance, the alligator harvest in the Southeast during the late 19th and early 20th centuries resembled what economists call open-access resource use. Anyone with a boat and a rifle could hunt alligators. This led to widespread overhunting, especially as prices for alligator hides increased. By the mid-20th century, alligator populations had plummeted.

Early uncontrolled harvest and the economy around it

The industrial revolutions of the 19th and 20th centuries had a profound impact on the alligator harvest market, with the second industrial revolution (mid-1800s to early 1900s) playing a particularly significant role. Enhanced transportation networks, including canals, railways, and steamships, facilitated the efficient and cost-effective movement of large quantities of alligator hide from remote wetlands to urban centers and ports. This improved access expanded markets and lowered transportation costs.

At the same time, industrialization increased demand for leather products such as boots, belts, and saddles. Due to its durability and unique texture, alligator leather became highly sought after for manufacturing these goods. Rapid urbanization further contributed to market expansion, as growing cities like Jacksonville, Florida, became major hubs for alligator hide trade, serving both domestic and international markets.

Technological advances in hunting equipment also enabled hunters to capture alligators more efficiently and on a larger scale. These factors combined led to significant declines in alligator populations during this period.

Economic data from the late 1800s illustrates the scale of the trade. In Miami, alligator hides sold for around seven dollars per six-foot section. Historical records indicate that buyers in the Big Cypress area purchased as many as 5,000 hides per month, often sourced from

Native American hunters. In one notable case, a single trapper was contracted to deliver 5,000 hides to a Paris leather company. Louisiana's alligator hide trade was similarly substantial, with approximately 3.5 million hides harvested between 1880 and 1933. This era represented the peak of commercial alligator hunting in the United States, particularly in states such as Louisiana and Florida.

Overharvest and Institutional Changes: Applying Gordon Schaeffer and Lueck's Model

Open Access and Overharvest: The Gordon–Schaefer Model

The Gordon-Schaefer model is a foundational framework in renewable resource economics that combines biological population growth with economic harvesting behavior. Developed in the mid-20th century by H. Scott Gordon and M. B. Schaefer, it was originally designed for fisheries but has since been widely applied to renewable natural resources such as wildlife populations.

The model assumes the population of a renewable resource, such as fish or alligators, grows according to a logistic growth function:

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right) - H$$

where:

P is the population stock,

r is the intrinsic growth rate,

K is the environmental carrying capacity,

H is the harvest (extraction) rate.

The harvest function is proportional to both effort and population size:

$$H = q \times E \times P$$

where:

q is the catchability coefficient (harvest efficiency),

E is harvesting effort (e.g., number of hunters or traps),

P is the population stock.

Equilibrium occurs when population growth balances harvest, such that stock remains constant:

$$rP\left(1 - \frac{P}{K}\right) = qEP$$

The model then identifies sustainable harvest levels and effort that maximize economic returns without depleting the resource.

A critical insight of the Gordon–Schaefer model is the outcome of open access conditions when no party holds exclusive rights, or regulatory enforcement reduces harvest effort. Under these circumstances, individuals maximize short-term profits by increasing effort, ignoring the collective impact on population. This leads to excessive harvesting, stock depletion, and eventual economic rent dissipation, embodying the classic "tragedy of the commons."

Defining property rights or regulatory controls shift incentives toward sustainable harvesting by making resource ownership economically valuable.

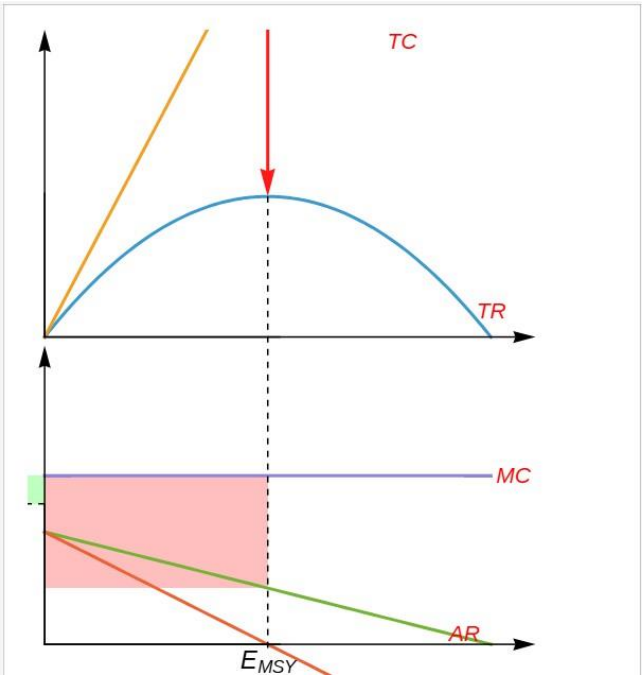
Importantly, while the model uses a logistic growth function, it is fundamentally a static equilibrium model. Gordon's original conception focused on equilibrium stock levels rather than explicitly modeling population changes over time. It provides snapshots where growth and harvest are instantaneously balanced, lacking explicit temporal dynamics that describe how populations decline or recover step-by-step.

This simplification means the model does not simulate:

- Population fluctuations or lagged recovery,
- Stochastic environmental effects,
- Complex feedback or multi-species interactions.

In the late 19th and early 20th centuries, the American alligator population faced conditions closely resembling the open-access scenario described by the model. Lack of enforceable ownership or regulation created a legal vacuum where unrestricted hunting prevailed. Harvesters increased effort as populations declined, leading to collapse consistent with the Gordon–Schaefer prediction of rent dissipation.

However, the real-world alligator population exhibited complex temporal dynamics not captured directly by the model's static framework. The decades-long decline and eventual recovery required integration of empirical population monitoring and adaptive governance beyond the model's scope.



The graph above illustrates two fundamental relationships in the Gordon–Schaefer bioeconomic model. It is a depiction of extreme conditions which led to the decline of the American Alligator Population.

The top panel of the diagram presents total revenue (TR) and total cost (TC) as functions of harvesting effort. The TR curve rises initially as greater effort results in increased harvest and revenues but eventually declines due to overexploitation and resulting reductions in the resource stock. The TC line, by contrast, is linear, reflecting the assumption of constant cost per unit of harvesting effort. The point at which TR reaches its maximum corresponds to the maximum sustainable yield (MSY), the biological optimum where the resource can produce the highest possible continuous yield without depleting its population. However, beyond this point, further increases in effort lead to lower harvests and declining revenue as the resource base becomes degraded.

The bottom panel further explores the relationship between effort, costs, and revenues. It typically depicts average revenue (AR), marginal cost (MC), and possibly marginal revenue as well. The intersection at EMSY highlights the level of effort associated with MSY. Importantly, the area between average revenue and marginal cost for effort levels below EMSY represents the possible economic profit available when management controls effort below open-access levels.

In the historical context of American alligator management, the industry initially operated as an open-access resource. During this period, individual hunters increased their effort without restriction in pursuit of private gain, causing the aggregate effort to approach or even exceed the open-access equilibrium where total revenue equals total cost and no economic rent remains. This scenario led to biological overexploitation and severely diminished alligator populations, as the Gordon–Schaefer model predicts.

Subsequent regulatory interventions, including the establishment of quotas, licensing systems, and landowner-based harvest allocations, harvesting effort leftward on the graph toward levels consistent with maximum economic yield (MEY), where profits and sustainability are maximized. This moves away from open access allowed the alligator

population to recover and ensured ongoing economic value from the resource.

It is important to note that the Gordon–Schaefer model, as depicted in these graphs, is based on static equilibrium analysis. It identifies long-run relationships between effort, revenue, and cost, but does not explicitly model the year-to-year population changes and delayed effects observed in real-world management, such as during the alligator's population collapse and subsequent recovery. Thus, while the model provides valuable insight into the risks of open access and the benefits of regulated management, dynamic ecological and historical data are necessary to fully understand the actual patterns of decline and restoration in the American alligator population.

Institutional Collapse and Reform: Lueck's Model

Indigenous tribes such as the Seminole, Choctaw, and Creek traditionally governed alligator use through informal but effective common property systems. Access to wetlands and wildlife was limited to tribal members, harvesting was regulated by cultural norms and seasonal knowledge, and overexploitation was discouraged through kinship-based rules and subsistence-focused practices (Hudson 1976; Goss 2010). These systems maintained ecological balance and prevented resource depletion, even in the absence of formal legal institutions.

However, the onset of colonization and the legal displacement of tribal authority created a major institutional shift. The breakdown of Indigenous governance left formerly regulated territories without clear property rights or regulatory oversight. In Lueck's (2002) terms, this created an "institutional vacuum" that transformed the wetlands into an open-access resource. Commercial hunters, driven by the high value of alligator hides, entered these lands without restriction or long-term responsibility. The state had yet to establish conservation policies, and no formal institutions had replaced tribal regulation.

This vacuum closely aligns with the dynamics predicted by both Lueck's framework and the Gordon–Schaefer model. With high market prices and no enforcement of access limits, individual hunters increased their harvesting effort to maximize short-term gains. In doing so, they dissipated economic rents and accelerated population decline. This race to extract value, typical of open-access conditions, led to the overharvesting crisis that pushed the alligator population toward the brink of collapse in the mid-20th century.

Efforts to monitor and manage alligator populations during this period were also constrained by ecological and logistical challenges. Alligators inhabit dense, swampy environments, exhibit nocturnal behavior, and often remain submerged, making them difficult to observe and count accurately. Early survey methods were inconsistent and often unreliable. Environmental variation, technological limitations, and mistrust from local communities further hindered data collection. These obstacles made it difficult for state agencies to respond effectively to population declines, delaying the development of sustainable governance strategies.

The turning point came in the late 20th century, when formal institutions began to fill the governance vacuum. States and federal agencies introduced policies such as permit systems, harvest quotas, egg collection regulations, and tagging programs. These mechanisms created property-like rights that limited access, clarified enforcement, and enabled long-term resource planning. By assigning responsibility and rights to specific users—farmers, licensed hunters, and landowners—these reforms realigned incentives toward conservation. Stakeholders now had a vested interest in preserving alligator populations, as the benefits of stewardship were internalized.

Lueck's model provides a strong theoretical foundation for understanding both the collapse and recovery of the American alligator. It shows how the transition from an informal common property system to an institutional vacuum, and finally to a structured regulatory

framework, shaped economic behavior and biological outcomes. The case illustrates the broader principle that sustainable resource use depends not only on ecological factors, but also on the presence of effective and enforceable governance institutions. Indigenous tribes such as the Seminole, Choctaw, and Creek traditionally governed alligator use through informal but effective common property systems. Land and wetlands were communally held, and access was limited to tribal members. Harvesting was regulated by cultural norms, seasonal knowledge, and kin-based enforcement. These systems, while not legally formalized, created clear property rights within the tribe and supported sustainable resource use.

With colonization, these governance systems broke down. Tribal land tenure was legally dismantled, and authority over resource use was removed. As a result, large areas of wetland habitat entered what Lueck (2002) describes as an institutional vacuum. There were no clear private or public property rights, and no effective enforcement of access or harvest limits. Commercial hunters, facing no restrictions, treated the land and alligator populations as open-access resources.

During this period, the state had not yet established regulations, and landownership in many wetland areas remained unclear or unenforced. Hunters entered these areas freely, targeting alligators for their high-value hides. This led to a race to harvest, with short-term profit motives outweighing any concern for conservation. The incentives in this setting align with Lueck's model and the Gordon–Schaefer framework: rising market prices, increasing harvest effort, and declining populations.

At the same time, efforts to monitor alligator numbers were limited by environmental and practical challenges. Alligators live in dense, swampy habitats, are mostly nocturnal, and often remain submerged. Early survey methods were unreliable, and state agencies lacked the tools, data, and trust needed to manage populations effectively. As a result, overharvesting continued unchecked for decades.

This changed in the late 20th century, when formal governance structures were introduced. Federal and state governments implemented permit systems, harvest quotas, and tagging programs. In some cases, private landowners were given exclusive rights to collect eggs or harvest alligators on their land. These policies replaced open access with enforceable rights and responsibilities. By tying access to landownership or permits, the state aligned economic incentives with conservation outcomes.

Lueck's framework helps explain both the collapse and recovery of the American alligator. The initial decline followed the breakdown of common property governance and the absence of clear landownership or regulation. The recovery came when institutions reestablished control over both land and resource use. This case shows how formal property rights and governance are essential for managing wildlife sustainably.

1900s - Accelerated hunting and shifts in the market

Between 1900 and 1947, the American alligator went through a major decline due to commercial overhunting. For most of this time, there were no regulations to oversee alligator hunting, so overharvesting was a common practice. This period also saw huge shifts in how alligator hides were used and valued, with both booms and crashes in the market. Eventually, by the 1940s, people started to realize that the species' population was endangered, and Florida became the first state to pass protective laws.

In the early 1900s, the alligator hide business was booming. States like Louisiana and Florida were leading the way, but alligator hunting was happening across much of the South. By 1902, over 140,000 alligator skins were being processed by U.S. tanneries each year, and Florida and Louisiana made up almost half of that (Hines 1979). Things took off in the 1920s, especially in 1929, when Florida alone produced about 190,000 hides (Allen and Neill 1949).

But the market wasn't always steady. In 1912, tanneries suddenly stopped buying hides, probably because there were too many on the market or the leather quality wasn't good enough

(Hines 1979). Demand picked up again in the late 1920s, thanks to better tanning techniques and a rise in popularity for alligator leather in things like bags, belts, and shoes (PanAm Leathers 2012).

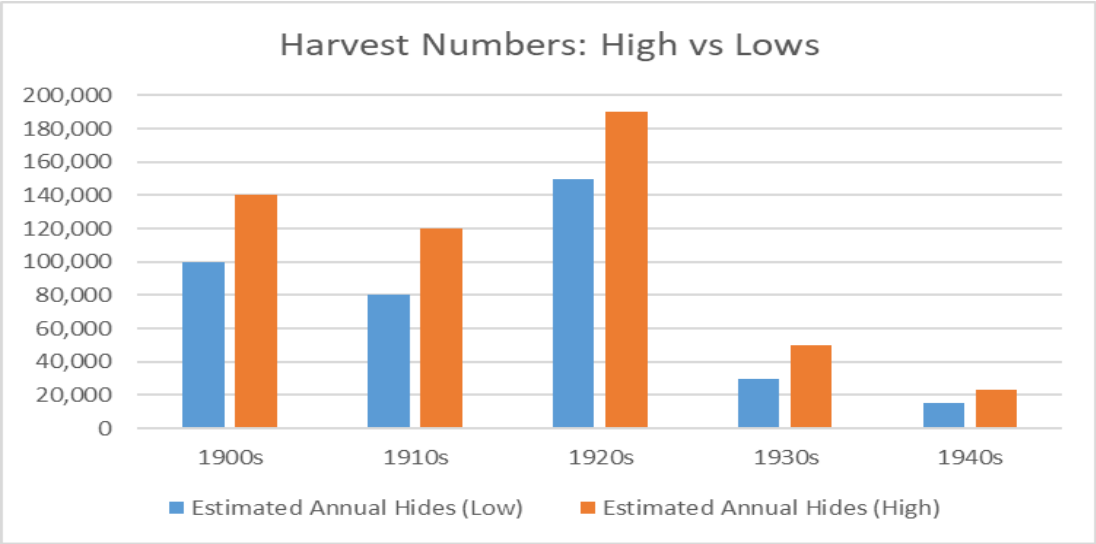
The Great Depression in the 1930s hurt trade again. People weren't spending on fancy products, and during World War II, a lot of hunters left for the military, which lowered the number of hides being collected. But after the war, prices jumped fast. In 1947, a high-quality 7-foot hide could sell for over $13, which was the highest it had ever been (Hines 1979). Even though alligators were becoming rare, the money to be made from hides kept some hunters going.

For most of this period, there were no laws regulating alligator hunting. The federal Lacey Act of 1900 prohibited interstate trade of illegally obtained wildlife, but because no states had yet banned alligator hunting, it had little effect (U.S. Fish and Wildlife Service 1987). Florida broke new ground in 1943 by passing the first state-level regulation: a closed hunting season during the breeding period and a size minimum that prohibited the harvest of alligators under 4 feet. This was the first formal effort to prevent total population collapse (Hines 1979). Other

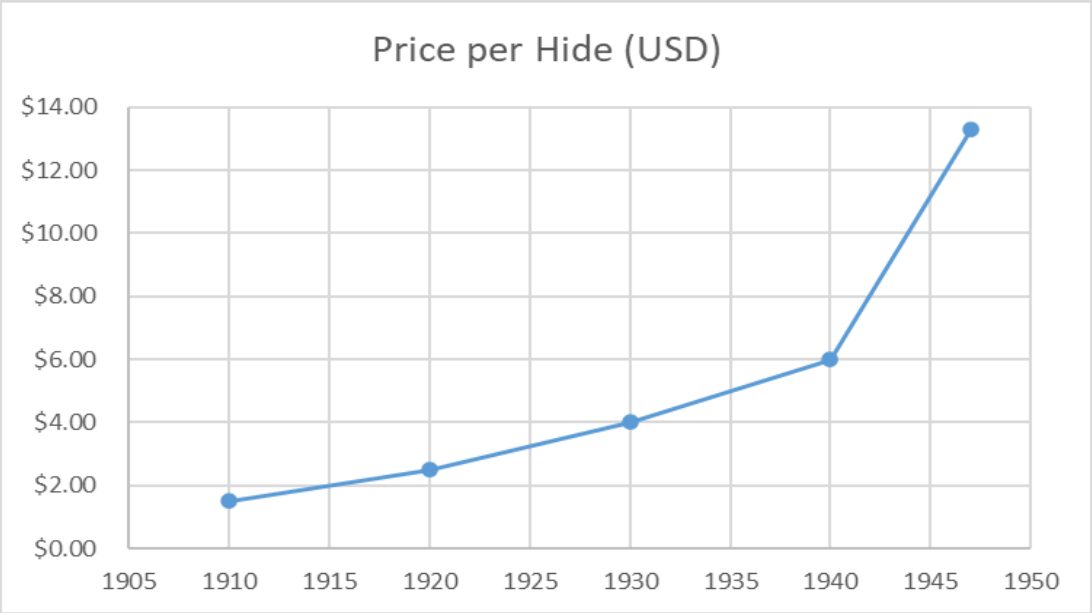
Alligator range states, including Louisiana, did not adopt similar restrictions until much later.

The consequences of open-access hunting were severe. Alligator populations across the Southeast declined dramatically. Large adults were specifically targeted, which interfered with reproduction and disrupted the age structure of wild populations (Joanen and McNease 1987). By the 1940s, alligators had disappeared from many areas where they were once abundant. In some parts of Louisiana, for example, alligators had been almost completely extirpated by the late 1940s.

Florida's 1943 law may have helped slow the decline locally, but populations remained extremely low throughout the region. Wildlife officials began to express concern that, without more widespread legal intervention, the species could disappear entirely from parts of its historic range.



Graph of the highest and the lowest harvest numbers in each decade from 1900 to the 1940s



Price fluctuations in the period 1910-1945

Years	**Estimated Harvest Volume (Southeast)**	**Avg Price per Hide (USD)**
1900	50,000	3.5
1910	70,000	2.8
1920	120,000	3.2
1925	160,000	3
1929	190,000	2.5

State Focus - Diving into Alligator populations of Florida and Louisiana

In the early 20th century, Florida and Louisiana treated the alligator as a commercially valuable but unregulated resource. Throughout the 1900s to the 1930s, both states experienced intense overharvesting driven by the leather trade. In 1929, Florida alone exported around 190,000 skins, one of the highest annual harvests on record (Hines 1979).

Louisiana's marshes were also overexploited similarly, with an estimated 3.5 million skins removed from 1880 to 1933 (Elsey, Joanen, and McNease 2021). This was one of the most primitive examples of the effects of the open-access model, where hunters faced very few legal liabilities while getting a substantial return through increased harvesting efforts until the population collapsed.

In the 1940s, population decline had become more prominent. Florida passed its first major policy in 1943: banning hunting during the breeding season and limiting the size of animals taken (Hines 1979). While this was an early step, enforcement remained weak and uneven. Louisiana lagged slightly behind Florida in adopting conservation measures during this

period. The open-access conditions largely continued in Louisiana until the early 1960s, when populations had declined so severely that regulatory involvement became unavoidable.

Below is a table showing the trend of alligator populations in Florida and Louisiana

Year	**Florida Population**	**Louisiana Population**
1950	500,000	800,000
1960	300,000	400,000
1970	200,000	250,000
1980	400,000	600,000
1990	700,000	1,000,000
2000	1,000,000	1,800,000
2010	1,200,000	2,500,000
2020	1,300,000	2,900,000
2024	1,300,000	2,900,000

Major shifts in institutional regulation began in the 1960s. 1962 marked the beginning of complete bans on alligator hunting. This marked a turning point in the protection of wild alligators. Enforcement was challenging at first-poaching and illegal interstate hide sales persisted through the mid-1960s (U.S. Fish and Wildlife Service 1987). However, the 1969 amendment to the Lacey Act and the listing of the American alligator as endangered in 1967 under federal law provided stronger regulatory tools. These laws prohibited interstate trade in illegally harvested alligator hides, which helped reduce market-induced poaching (U.S. Fish

and Wildlife Service 1987; Hines 1979).

The 1970s saw the emergence of a more nuanced policy approach. Louisiana moved first by launching a carefully monitored experimental harvest in Cameron Parish in 1972. Licensed hunters were issued tags, and the harvest was confined to specific areas that showed positive signs of population recovery (Elsey, Joanen, and McNease 2021).

Florida, on the other hand, continued to ban hunting but introduced a nuisance removal program in 1978 that allowed licensed trappers to kill dangerous alligators and sell the meat and hides. This program provided both public safety benefits and limited commercial incentives without resuming general public hunting (Florida Fish and Wildlife Conservation Commission 2021).

Farm vs Wild Harvesting

Up until the 1960s, alligators across the South were hunted heavily for their hides, leading to population collapse. The federal listing of the species as endangered in 1967 (later reclassified in 1987) allowed states like Florida and Louisiana to ban hunting and begin recovery efforts (U.S. Fish and Wildlife Service 1987; Hines 1979). As populations recovered, state-managed hunting programs and commercial farming emerged to meet leather market demand while protecting wild stocks.

Presently, wild hunting is highly regulated. Florida issues alligator hunting permits via lottery, allowing about 7,000–8,000 gators to be harvested annually under quotas tied to regional population data (Florida Fish and Wildlife Conservation Commission 2021). The state also runs a year-round nuisance removal program to handle human-wildlife conflict, with trappers allowed to sell meat and hides from captured animals (Hines 1979).

Louisiana uses a private lands model: landowners receive a quota of tags based on aerial nest surveys and habitat quality. Hunters remove around 24,000 alligators per year, mostly over

a short season, and the system creates strong financial incentives to preserve wetlands (Elsey, Joanen, and McNease 2021).

Alongside hunting, alligator farming has become a major industry. In Louisiana, licensed farms collect eggs from wild nests, hatch them, and raise juveniles under controlled conditions.

Roughly 10% are released back into the wild at 3–4 feet, helping restock populations with individuals that now have high survival odds (Elsey, Joanen, and McNease 2021). The remaining 90% are processed for meat and hides. In 2019, Louisiana farms harvested over 438,000 alligators, generating more than $86 million in hide value alone. Florida's farms operate similarly but aren't required to return juveniles. Instead, they focus on raising gators from legally collected eggs and producing hides for the global luxury leather market (Florida Fish and Wildlife Conservation Commission 2021).

Economically, both systems benefit conservation. Wild harvest programs give landowners reasons to conserve habitat and fund wildlife agencies through tag and permit revenue. Farming reduces pressure on wild populations and supports jobs in rural areas. All hide trade is tracked through the CITES tagging system, and export requires federal permits (U.S. Fish and Wildlife Service 1987). Though animal welfare laws don't apply to reptiles federally, states like Florida and Louisiana regulate farm licensing and egg collection closely.

The Economics of Alligator Governance: 1940s to Present

In the mid-20th century, the American alligator faced near extinction as a result of unregulated hunting driven by high hide demand. In the absence of property rights or effective state intervention, alligators were harvested under open-access conditions, consistent with Gordon's (1954) bioeconomic model. In this model, renewable resources like alligators are subject to overexploitation when individuals harvest up to the point where average revenue equals marginal cost, leading to rent dissipation and eventual biological collapse. This dynamic played

out in real time during the 1940s and 1950s, as hide prices rose and populations plummeted (Hines 1979; Allen and Neill 1949)

The collapse prompted a shift to command-and-control governance in the 1960s. States like Louisiana and Florida banned alligator hunting, and the federal government listed the species as endangered in 1967. These bans eliminated market access and allowed populations to recover. However, without legal trade, economic value was removed from the species, and incentives for habitat conservation, especially on private land, declined. This created what Lueck (2002) calls a "conservation without markets" regime, where state agencies bore the full cost of protection without private participation.

The major economic transformation began in the 1970s and 1980s, as policymakers shifted toward market-based conservation. With populations rebounding, Louisiana pioneered a regulated wild harvest program in 1972, and Florida followed with public hunts and nuisance removal programs in the 1980s (Elsey, Joanen, and McNease 2021; Florida Fish and Wildlife Conservation Commission 2021). These programs introduced permits, quotas, and

landowner-based tagging systems—essentially converting access from open to quasi-privatized regimes. Landowners received tags tied to the habitat quality on their properties, giving them financial incentives to conserve wetlands rather than drain them for agriculture or development. The link between conservation and economic return created positive feedback loops: more habitats meant more tags, which meant more revenue from leasing land to hunters or collecting eggs for farms.

At the same time, alligator farming and ranching became economically viable. Louisiana's 1986 ranching model allowed farms to collect wild eggs under permit, raise the juveniles in captivity, and return a portion into the wild. This "HeadStart" program created a dual benefit: farms profited with high survival rates and rapid growth while wild populations received a consistent influx of juveniles that had skipped their most vulnerable life stage. The returns were substantial. In 2019, Louisiana farms generated over $86 million in hide value

alone (Elsey, Joanen, and McNease 2021). Private landowners also received per-egg payments from farmers, reinforcing habitat stewardship on working lands.

The economic sustainability of these governance models depends heavily on global hide prices and market demand. When alligator leather was in fashion, such as in the late 1980s and

mid-2000s, harvests and egg collections rose accordingly. But market crashes, like the one following the 2008 financial crisis, led to steep declines in hide prices (from over $40/foot to under $10), making it unprofitable for many hunters and farmers to participate (Chabreck et al. 2011). This boom-bust cycle reflects the economic risk of wildlife markets tied to luxury goods. However, because modern governance ties harvest levels to population monitoring, not price, biological sustainability has not been hampered.

Federal oversight under CITES and the Lacey Act also plays a major economic role. By requiring every harvested or farmed alligator to be tagged and tracked, U.S. authorities maintain trade legitimacy and deter black-market laundering of protected species. This traceability boosts consumer confidence in alligator products and ensures access to export markets, which are vital for farm profitability (U.S. Fish and Wildlife Service 1987).

In recent decades, governance has evolved to support both conservation and economic efficiency. States use adaptive management to adjust quotas and egg collection based on population surveys, following Ostrom's (1990) principles of effective resource governance. In Florida, permit fees fund monitoring and enforcement, while in Louisiana, industry participation in advisory councils ensures regulatory responsiveness. These measures have made the system economically sustainable while supporting long-term population recovery.

Conclusion

The American alligator underwent extensive population decline during the early 20th century as a result of unregulated, open-access hunting driven by rising hide prices and expanding market demand. In the absence of defined property rights or regulatory oversight, harvest levels exceeded sustainable limits, consistent with common-pool resource theory. Federal listing of the species under the Endangered Species Act in 1967, combined with state-level hunting bans and legal trade restrictions, led to the species' recovery beginning in the 1970s.

During the recovery period, governance frameworks evolved to incorporate regulated wild harvest and commercial alligator farming. States like Florida and Louisiana implemented quota-based hunting systems that linked allowable harvests to local population surveys. These systems used state-issued tags, permits, and season limits to control access and effort.

Concurrently, commercial farming and ranching operations were developed, particularly in Louisiana, where wild-collected eggs were reared in captivity and a portion of juveniles were returned to the wild to support population growth.

Land ownership emerged as a key factor in modern governance. In Louisiana, over 80% of coastal wetlands are privately owned. The state's harvest and ranching programs assign tags and egg collection rights to landowners based on habitat quality and nesting density, creating direct financial incentives to conserve alligator habitat. These programs converted wetlands into productive assets by aligning landowner income with the ecological value of their property. In Florida, both public and private lands are used for farming and hunting, with state agencies

overseeing access through permit systems. Throughout the alligator's range, landowners who host nesting habitat now receive payments or market access in exchange for maintaining ecological conditions conducive to alligator reproduction.

Federal oversight remains integral to this system. All legal trade is subject to CITES tagging requirements and compliance under the Lacey Act, ensuring that farmed and wild-harvested alligators are tracked from source to export. The combination of land-based incentives, biological monitoring, and trade regulation has supported the development of a structured, sustainable use regime.

Today, the American alligator is considered biologically secure across its native range. It is managed as a renewable resource under state and federal law, with wild and farmed harvests governed by population data, property rights, and market structures. Conservation outcomes are maintained through a governance framework that integrates ecological monitoring with economic incentives tied to land use and regulated trade.

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Excel Sheet used for data analysis American Alligator Data.xlsx

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